



Human-Centric Artificial Intelligence Pedagogy (HCAP) framework developed from TPACK through integration of artificial intelligence literacy and competency

Thomas K.F. Chiu

To cite this article: Thomas K.F. Chiu (15 Jan 2026): Human-Centric Artificial Intelligence Pedagogy (HCAP) framework developed from TPACK through integration of artificial intelligence literacy and competency, Interactive Learning Environments, DOI: [10.1080/10494820.2026.2615818](https://doi.org/10.1080/10494820.2026.2615818)

To link to this article: <https://doi.org/10.1080/10494820.2026.2615818>



© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group



Published online: 15 Jan 2026.



[Submit your article to this journal](#)



Article views: 6709



[View related articles](#)



[View Crossmark data](#)



Citing articles: 9 [View citing articles](#)

Human-Centric Artificial Intelligence Pedagogy (HCAP) framework developed from TPACK through integration of artificial intelligence literacy and competency

Thomas K.F. Chiu  a,b,c

^aDepartment of Curriculum and Instruction, The Chinese University of Hong Kong, Hong Kong, People's Republic of China; ^bCentre for Learning Sciences and Technologies, The Chinese University of Hong Kong, Hong Kong, People's Republic of China; ^cCentre for University and School Partnership, The Chinese University of Hong Kong, Hong Kong, People's Republic of China

ABSTRACT

The rise of artificial intelligence (AI) in education, particularly generative AI, challenges the sufficiency of the established Technological Pedagogical Content Knowledge (TPACK) framework. AI's agentic autonomy, epistemic complexities, and ethical dimensions necessitate an evolved model. This study investigates the newly proposed Human-Centric AI Pedagogy (HCAP) framework, designed to address these gaps by integrating five knowledge domains: AI-Technological, AI-Content, AI-Pedagogical, Human-AI Collaborative, and Ethical Knowledge. The research goal was to define the specific knowledge and skills required within these HCAP domains from a teacher's perspective. Utilizing a three-round Delphi method, a panel of 30 teachers from diverse subjects developed a consensus list of essential competencies. The findings identified and refined 25 critical knowledge items, providing a foundational and empirically grounded model for the HCAP framework. These findings offer a concrete roadmap for teacher education, translating a theoretical model into actionable competencies. This study equips educators to transition from merely using AI to strategically orchestrating human-AI collaborative learning, ensuring they can harness AI's potential ethically, critically, and productively. The HCAP and its suggested knowledge serve as a vital tool for developing future-ready teacher training programs and professional development.

ARTICLE HISTORY

Received 16 November 2025
Accepted 7 January 2026

KEYWORDS

Human-centric AI pedagogy (HCAP); TPACK; artificial intelligence competency; artificial intelligence literacy; teacher education

Introduction

Artificial intelligence (AI) fundamentally reshapes contemporary educational landscape, especially with tools like generative AI models such as ChatGPT, DeepSeek, Gemini and adaptive learning platforms (Karataş & Ataç, 2025; Wong, 2025). This change is a major shift that calls into question whether the well-established Mishra and Koehler (2006)'s Technological Pedagogical Content Knowledge (TPACK) framework is still adequate. TPACK provides a foundational framework for analysing the knowledge required by educators to successfully incorporate technology into their instructional practices (Mishra & Koehler, 2006; Pamuk, 2012). However, its underlying assumptions were developed during a period characterized by mainly passive and deterministic digital tools, such as presentation software or instant response systems, which functioned in a predictable manner according to user commands (Chiu, 2025; Mishra et al., 2023).

Therefore, AI represents not merely an incremental technological advance, but a paradigm shift that challenges the core sufficiency of the traditional TPACK framework (Karataş & Ataç, 2025; Mishra et al., 2023). AI systems exhibit agentic autonomy, capable of generating original content, providing personalized feedback, and making independent decisions based on data (Chiu, 2024; Hughes et al., 2025). This autonomy transforms the technology from a passive instrument into an active, often unpredictable, participant in the teaching and learning process. This new process exposes critical gaps in TPACK, which can be categorized into four key areas. First, tool agency: understanding AI's operational logic, probabilistic nature, and the skills required

CONTACT Thomas K.F. Chiu  Thomas.kf.chiu@gmail.com, tchiu@cuhk.edu.hk

This article was originally published with errors, which have now been corrected in the online version. Please see Correction (<http://dx.doi.org/10.1080/10494820.2026.2633838>)

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

to steer it, such as prompt engineering is not emphasized in TPACK (Celik, 2023; Celik et al., 2025; Chiu, 2025; Chiu et al., 2024). Second, epistemic challenges: AI's autonomous content creation demands teachers shift from content deliverers to critical validators, a competency not covered by TPACK (Aldemir et al., 2025; Chiu et al., 2024). Third, ethical complexity: AI makes ethical reasoning a core, explicit domain of teacher knowledge, whereas it was often implicit in TPACK (Aldemir et al., 2025; Mishra et al., 2023). Finally, dynamic collaboration: TPACK lacks a dedicated knowledge domain for conceptualizing the move from human-technology interaction to human-AI collaboration, where cognitive tasks are shared between teachers, students, and AI agents (Aldemir et al., 2025; Chiu et al., 2024).

In direct response to these profound limitations, scholars have called for a reconceptualization of TPACK tailored for the age of AI (Chiu, 2025; Celik, 2023; Mishra et al., 2023). This has led to Chiu (2025)'s proposal of the Human-Centric AI Pedagogy (HCAP) framework. Human-Centric AI Pedagogy (HCAP) expands and redefines the original model by integrating AI literacy and competency as foundational pillars (Zhou, Li, et al., 2025). It introduces five interdependent knowledge domains: three that transform the original TPACK cores through a sociotechnical lens – I-Technological Knowledge (I-TK), AI-Content Knowledge (I-CK), and I-Pedagogical Knowledge (I-PK) – and two entirely novel domains: Human-AI Collaboration Knowledge (HAIC-K) and Ethical Knowledge (Ethics-K). While Human-Centric AI Pedagogy (HCAP) provides a crucial theoretical advancement for reimagining teachers as designers and conductors of human-AI collaborative learning, a significant research gap persists. The framework remains a high-level conceptual structure and does not yet clearly define the specific, actionable knowledge and skills required within each of its five domains for effective classroom implementation.

Therefore, this study aims to operationalize the Human-Centric AI Pedagogy (HCAP) framework by identifying its core knowledge and skills. The primary research goal is to answer the question: What knowledge and skills are required for the five domains of HCAP from a teacher's perspective? Given that teacher education for AI is still in its infancy and remains largely exploratory, a consensus on these necessary competencies has not yet been established. This study uses Delphi method to gather and refine expert consensus from practicing teachers, thereby translating the theoretical HCAP framework into a concrete and validated set of competencies. The findings are intended to provide a foundational roadmap for developing targeted teacher training programs, designing effective professional development, and guiding future empirical research. The suggestions empower educators to harness the transformative potential of AI in an ethical, healthy, and productive manner.

Theoretical background

TPACK framework and its insufficiency in the AI era

The TPACK framework, introduced by Mishra and Koehler (2006), has served as a seminal model for understanding teacher knowledge required for effective technology integration. TPACK posits that successful technology-enhanced learning is not merely about knowledge of technology (TK), pedagogy (PK), or content (CK) in isolation, but resides in the complex interplay between these three core components. The framework emphasizes the importance of Technological Pedagogical Knowledge (TPK) – understanding how teaching and learning change with technology; Technological Content Knowledge (TCK) – comprehending how technology and content influence each other; and Pedagogical Content Knowledge (PCK) – the core of teaching how to make a subject understandable. For over a decade, TPACK has provided researchers and practitioners with a valuable lens for diagnosing teacher readiness and designing professional development for a wide array of digital tools, from interactive whiteboards to productivity software (Pamuk, 2012).

However, TPACK was developed in the period of non-generative technology that is less dynamic, personalized and interactive (Aldemir et al., 2025; Celik, 2023; Celik et al., 2025; Chiu et al., 2024; Karataş & Ataç, 2025; Mishra et al., 2023). The advent of generative and adaptive AI technologies – such as ChatGPT, Mid-journey, and intelligent tutoring systems – has exposed critical limitations in the TPACK framework (Chiu et al., 2024; Karataş & Ataç, 2025). AI is not just another tool in the educational toolkit; it is an agentic, generative, and socio-ethically complex entity. Its unique characteristics challenge the foundational assumptions of TPACK, revealing its insufficiency in preparing teachers for the realities of the AI-driven classroom. This insufficiency is evident across its core domains and is most acute in two areas it scarcely addresses: human-AI collaboration and embedded ethics.

The core domains of the TPACK framework – TK, CK, and PK – each reveal unique inadequacies when considered in light of the challenges posed by AI. Traditional TK is centered on the operational use of relatively non-generative tools, such as spreadsheets or presentation software. However, in the AI context TK requires a deeper understanding of AI's operational logic, including prompt engineering, algorithmic awareness, and system auditing. This means teachers must be able not only to use technology, but also to structure queries to large language models to avoid hallucinations and critically assess output reliability. Meanwhile, CK in TPACK assumes the teacher is the primary authority on content, yet AI's generative capabilities fundamentally disrupt this dynamic; teachers now need additional CK that involves critically curating, validating, and contextualizing AI-generated learning materials. Teachers should have stronger epistemic beliefs – awareness and vigilance, to triangulate AI outputs against trusted sources, identify inaccuracies and ensure generated content aligns with curricular standards. Their roles move from content deliverers to content validators. Finally, PK traditionally involves designing learning experiences around inert tools, but with AI acting as an active participant, teachers must be capable of collaborating with AI to provide a more personalized learning environment, and redesigning assessments approaches. Although the evolution of TK, CK, and PK remains essential, the most significant limitations within TPACK are found in its marginal consideration of human-centred learning (Chiu, 2024; UNESCO, 2024) in the context of AI. These gaps can be categorized into four key areas: tool agency, epistemic challenges, ethical complexity, and dynamic collaboration.

Tool agency: Traditional digital tools are passive instruments; they do what the user commands. In contrast, AI systems exhibit agentic autonomy, capable of generating original content, providing personalized feedback, and making independent decisions based on data (Hughes et al., 2025). This autonomy means the technology is no longer just a tool but an active participant in the educational process. TPACK's TK, which focuses on the operational use of static tools, is inadequate for understanding AI's operational logic, its probabilistic nature, and the skills required to steer it effectively, such as prompt engineering and system auditing (Aldemir et al., 2025; Celik et al., 2025; Karataş & Ataç, 2025).

Epistemic challenges: AI's capacity for autonomous content creation introduces a profound epistemic challenge (Vučković & Sikimić, 2025). Teacher's role shifts from a content deliverer to a critical validator and contextualizer (Mishra et al., 2023). Traditional Content Knowledge (CK) does not prepare educators for the task of critically curating, triangulating, and validating AI-generated outputs, a competency termed AI-Content Knowledge (I-CK) (Chiu, 2025). This requires epistemic vigilance to identify algorithmic biases, hallucinations, and inaccuracies, ensuring AI serves as a validated resource rather than an unchallenged authority.

Ethical Complexity: The ethical dimensions of AI are not peripheral concerns but central to its operation. Issues of data privacy, algorithmic bias, fairness, and accountability are inherent in AI systems (Vučković & Sikimić, 2025). While ethics may have been an implicit consideration in TPACK, the pervasive and often opaque nature of AI necessitates that ethical reasoning be a core, explicit domain of teacher knowledge (Aldemir et al., 2025; Karataş & Ataç, 2025). The risks of exacerbating societal inequalities, spreading misinformation, and eroding student critical thinking are too significant for ethics to remain an afterthought (Chiu, 2025).

Dynamic Collaboration: The most significant shift may be the move from human-technology interaction to human-AI collaboration (Chiu & Rospigliosi, 2025). The human-AI collaboration model posits that learning efficacy is maximized when teachers, students, and AI coordinate cognitive tasks (Wen et al., 2025). This requires formalizing knowledge on how to structure interactions where AI handles procedural tasks (e.g. grammar correction, data analysis), thereby freeing teachers for higher-order mentorship, creative facilitation, and socio-emotional support. TPACK lacks a dedicated knowledge domain to conceptualize and orchestrate these dynamic, collaborative workflows.

Overall, an advancement of TPACK is therefore necessary to empower teachers to navigate this new complex landscape. This advancement should incorporate AI-related knowledge and skills, including AI literacy and competency.

Defining AI literacy and competency in education

The integration of AI in mainstream education has given rise to two terms: AI literacy and AI competency. These terms are frequently used interchangeably within the educational sector, leading to a blurred understanding of AI-related knowledge and skills (Long & Magerko, 2020). The concept of AI literacy is grounded in

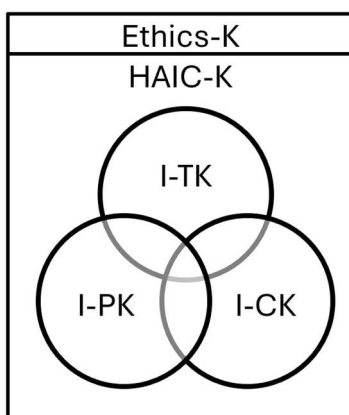
earlier frameworks of various literacies including mathematical, digital, computational, media, and language literacies (Heung et al., 2026). Long and Magerko (2020) provided one of the first and most widely cited definitions, framing AI literacy as “a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool online, at home, and in the workplace” (p. 2). Although this definition established an important basis, it introduced terminological ambiguity by referring to competencies in the context of literacy. Later studies such as UNESCO (2024) have primarily centered on this conceptual foundation, aiming to delineate the key knowledge and abilities necessary for foundational AI understanding. Common components of AI literacy include (i) conceptual understanding: Knowing what AI is, its history, and its core concepts like machine learning, neural networks, and data dependency; (ii) application and use: Understanding how AI is applied in various domains and developing the skills to use AI tools effectively; (iii) critical evaluation: developing the ability to assess AI outputs for reliability, bias, and appropriateness; (iv) ethical and societal implications: recognizing the potential benefits and risks of AI, including issues of privacy, fairness, and employment (Heung et al., 2026)

Human-Centric Artificial Intelligence Pedagogy (HCAP) framework in this study

Scholars have called for a reconceptualization of TPACK that specifically addresses the unique capabilities and challenges of AI (Aldemir et al., 2025; Celik, 2023; Celik et al., 2025; Chiu, 2025; Karataş & Ataç, 2025). Building on the foundations of sociotechnical pedagogy, human-AI collaboration models, and the concept of the responsible digital citizen, the HCAP framework has been proposed as a critical evolution see Figure 1. HCAP expands and redefines the original framework by integrating AI literacy and AI competency as foundational pillars, positioning teachers as augmented pedagogy designers who highlight human-AI centred learning.

The HCAP framework introduces five interdependent knowledge domains. First, the original TPACK domains are transformed through a sociotechnical lens: I-Technological Knowledge (I-TK) replaces TK, encompassing prompt engineering, algorithmic awareness, and system auditing. Intelligent-Content Knowledge (I-CK) supersedes CK, focusing on the critical curation and validation of AI-generated disciplinary content. AI-Pedagogical Knowledge (I-PK) transforms PK, involving the strategic design of AI-augmented learning experiences, including assessment redesign for skills AI cannot replicate, like creativity and critical thinking.

By focusing on human-AI centred learning, HCAP introduces two novel domains. Human-AI Collaboration Knowledge (HAIC-K) formalizes the understanding of how to structure productive interactions between humans and AI, defining roles for AI such as co-author, peer reviewer, or simulation partner (Chiu &



Note:

I-TK: Intelligent technological knowledge

I-PK: Intelligent pedagogical knowledge

I-CK: Intelligent content knowledge

HAIC-K: Human-AI collaboration knowledge

Ethics-K: Ethical knowledge

Figure 1. Human-Centric Artificial Intelligence Pedagogy (HCAP) framework in this study.

Rospigliosi, 2025). Most critically, Ethical Knowledge (Ethics-K) is centralized as a core domain that permeates all others, providing the normative frameworks for bias detection, fairness protocols, and responsible implementation. This embedded ethics ensures that tool auditing (I-TK), content validation (I-CK), and pedagogical design (I-PK) are consistently guided by moral and equitable principles.

The central idea of HCAP lies in the dynamic interdependence of these domains (Chiu, 2025). For instance, understanding model limitations (I-TK) is a prerequisite for validating AI-generated content (I-CK). Similarly, pedagogical designs (I-PK) determine how AI integrates into group dynamics (HAIC-K), while ethical knowledge (Ethics-K) governs the entire process, from the initial audit of an AI application to the design of equitable assessments. This integrated structure directly resolves the four key limitations of traditional TPACK by addressing AI's autonomy, its epistemic challenges, its inherent ethical complexity, and its role as a collaborative agent. The following describe each knowledge domains in the HCAP.

Intelligent technological knowledge (I-TK)

- Rooted in AI Literacy: This domain moves beyond basic digital literacy to a deep understanding of AI systems. It includes knowledge of core AI concepts (e.g. machine learning, neural networks), the ability to discern AI-powered tools from non-AI tools, and an understanding of their capabilities and limitations.
- Elevated by AI Competency: I-TK is demonstrated through practical, competent use. This includes prompt engineering (strategically crafting inputs to guide AI outputs), algorithmic awareness (understanding how data biases affect results), and system auditing (evaluating the reliability and suitability of an AI tool for a specific educational task). A teacher with I-TK does not just use an AI essay grader; they competently assess its grading criteria and calibrate it for their classroom context.

Intelligent content knowledge (I-CK)

- Rooted in AI Literacy: I-CK involves understanding how AI generates, represents, and transforms disciplinary content. It includes knowing the sources of AI training data and recognizing common patterns or tropes in AI-generated content (e.g. the formal tone of ChatGPT).
- Elevated by AI Competency: This domain is defined by a teacher's competency to act as a critical validator and contextualizer. A teacher with strong I-CK doesn't accept AI-generated content at face value. They cross-reference it with authoritative sources, identify potential inaccuracies or oversimplifications, and contextualize the information within the curriculum and real-world relevance. They shift from being the primary source of content to being the lead curator and critic of AI-generated knowledge.

Intelligent pedagogical knowledge (I-PK)

- Rooted in AI Literacy: I-PK is the knowledge of how AI can support teaching and learning processes. It involves understanding different AI-based learning platforms and their general pedagogical applications (e.g. adaptive tutors, AI brainstorming assistants).
- Elevated by AI Competency: Competency in I-PK is demonstrated through the strategic orchestration of AI-augmented learning experiences. This includes designing lessons that leverage AI for personalized scaffolding, rethinking assessment to measure human-centric skills (e.g. creativity, critical evaluation of AI outputs), and facilitating activities where students interact with AI in structured, learning-focused ways. It is the competency to seamlessly blend human instruction with AI support.

Human-AI collaboration knowledge (HAIC-K)

- Rooted in AI Literacy: This is the understanding that AI can be a collaborative partner. It includes (i) knowing that AI can take on different roles (e.g. tutor, coach, co-creator); (ii) recognizing the goal of human-AI collaboration is to augment, not replace, human intelligence.

- **Elevated by AI Competency:** This is the skillful facilitation and choreography of productive human-AI partnerships. It includes (i) meaningful role definitions: Structuring interactions by assigning specific, meaningful roles to the AI, Socratic questioner, data analyzer, simulation engine and complementary roles to the students (meaningful role definitions); (ii) agency cultivation: Ensuring that students remain the directors of the learning process, making final judgments, and integrating AI's contributions into their own original thinking and work; (iii) student originality: Creating learning activities to help individual students keep their original ideas using an iterative process of initial-ideas, co-creation, critique, and refinement.

Ethical knowledge (Ethics-K)

- **Rooted in AI Literacy:** This involves knowing the key ethical issues surrounding AI, such as data privacy, algorithmic bias, copyright, and the potential for misinformation.
- **Elevated by AI Competency:** EK is the active and ongoing competency to identify, mitigate, and teach about ethical dilemmas. A teacher with EK does not just understand bias theoretically; they proactively audit the AI tools they use for biased outputs, establish clear classroom protocols for data privacy when using AI, and design lessons that teach students how to recognize deepfakes or ethical use AI for their own work. It is a core competence that informs all other domains.

The HCAP framework represents a necessary and timely response, bridging the gap between foundational AI literacy and applied AI competency. By reconceptualizing teachers as designers and conductors of human-AI collaborative learning, HCAP provides a comprehensive roadmap for empowering educators to harness the transformative potential of AI ethically, critically, and effectively, thereby preparing students for a future inextricably intertwined with intelligent technologies. HCAP provides a comprehensive roadmap for empowering educators to harness the transformative potential of AI ethically, critically, and effectively, thereby preparing students for a future inextricably intertwined with intelligent technologies.

Research gaps

As we discussed, although the TPACK framework has been a vital starting point for education in the digital era, the rise of calls for a substantial shift in how we view teacher knowledge. The HCAP framework can reimagine teachers as designers and creators and leaders in human-AI collaborative learning (Chiu & Rospigliosi, 2025); however, it does not clearly define the types of knowledge and skills required (Chiu, 2025). More empirical research is needed to clarify the knowledge and skills and necessary within the HCAP framework.

This study and method

Research goal

This study aims to suggest knowledge skills needed for the HCAP framework using the Delphi approach. The major research question is:

What knowledge and skills are required for the five domains of HCAP from a teacher's perspective?

These findings assist researchers and practitioners in more effectively replicating, implementing, and synthesizing research related to teacher education with AI, as well as in developing teacher education programs (both pre – and in-service) for AI.

Research design

As teacher education for AI is still in its infancy and exploratory, the ideas on the knowledge and skills needed to teach with AI may not reach a consensus yet. A Delphi study uses an iterative process of questionnaires interleaved with controlled feedback to obtain a reliable consensus view from a group of experts (Chiu, 2024). It is possible to obtain a wide variety of opinions for decision-making. The anonymous

process avoids conflicts between experts and encourages objective participation. Accordingly, the Delphi research design was selected as suitable for this study's main research question.

Three Delphi rounds are usually enough to create an equilibrium where further rounds do not affect findings dramatically for most studies (Ahmadi et al., 2023; Teixeira et al., 2020). To gather a broad range of informed perspectives on the knowledge and skills required for effective AI integration in teaching, we convened a panel of school teachers representing various subject areas. Each teacher must have at least one year of experience in designing and implementing AI-based learning activities.

There is currently no established minimum panel size for Delphi studies, as consensus regarding appropriate sample sizes has not been reached (Jorm, 2015). Previous studies focusing have used between 10 and 34 participants in their designs (Ahmadi et al., 2023; Teixeira et al., 2020). In order to mitigate potential attrition, this study invited and engaged 30 participants from 15 schools. Participants represented diverse teaching disciplines, including English (5), Chinese (5), Mathematics (5), Science (5), Humanities & Art (5), and Technology (5). The selected schools comprises various academic levels among students in Hong Kong SAR.

Research procedures

First, together with the teachers, we created the initial list of the requisite knowledge for teaching with AI. We compiled the lists with two experienced teachers using the definition of AI literacy and competency discussed in the previous section; the initial list had 20 knowledge items, see Table 1. We distributed the survey online for this study.

During Round One, teachers provided qualitative feedback on each knowledge label, including its name and description. They also evaluated the relevance of each knowledge and skill to HCAP, specifying the corresponding domain when applicable. At the end of the survey, teachers received a comprehensive list and were asked to identify any knowledge items that appeared to be duplicates. To ensure a more exhaustive compilation, teachers were invited to propose additional knowledge they considered missing from the list. Following each round of the Delphi process, the list was refined based on teacher input. During the process of item revision, the two teachers and the author collaboratively reviewed and discussed the proposed changes until consensus was reached. When recommendations required significant changes – such

Table 1. Major revision over the three Delphi rounds.

	Initial List	Round One	Round Two	Round Three
Intelligent Technological Knowledge (I-TK): Mastery of AI systems' operational logic, capabilities, and limitations				
#1	Prompt engineering and interaction		Prompt engineering	Consensus
#2	Algorithmic awareness and model limitation			Limitation awareness
#3	System auditing and evaluation			System auditing
#4	Data fundamentals	Consensus		
#5		Tool integration	Consensus	
Intelligent Content Knowledge (I-CK): The ability to critically curate, validate, and contextualize AI-generated disciplinary content.				
#6	Epistemic vigilance and critical validation	Critical validation		Source triangulation
#7	Bias and inaccuracy detection		Bias detection	Consensus
#8	Contextualization and alignment	Contextualization	Consensus	
#9	Understanding of AI's disciplinary approach		Disciplinary methodological critique	Consensus
Intelligent Pedagogical Knowledge (I-PK): The strategic design of AI-augmented learning experiences.				
#10	Metacognitive orchestration		reflective practice	Consensus
#11	Assessment redesign	Consensus		
#12	Scaffolded collaborative design	Scaffolded collaboration	Scaffolded co-agency design	Consensus
#13	Differentiation and personalization	Personalization	Consensus	
#14				Ethical integration
Human-AI Collaboration Knowledge (HAIC-K): Structuring effective interactions and defining roles for AI within learning.				
#15	Role definition and task allocation			Dynamic role allocation
#16	Interaction flow designer	Consensus		
#17	Critical dialogue and interdependence		Critical dialogue	Consensus
#18			Critical interdependence	Consensus
#19	Group dynamics collaboration	Consensus		
Ethical Knowledge (Ethics-K): The normative frameworks for responsible and equitable AI use in education.				
#20	Bias mitigation and fairness protocols		Equity auditing	Consensus
#21	Data privacy	Consensus		
#22		Usage protocols	Consensus	
#23	Societal impact	Consensus		
#24			Well-being advocacy	Consensus
#25	Inclusive design	Consensus		

as notably different function descriptions – the updated learning activity was classified as a new activity. Modifications suggested by teachers after each round led to further revisions, with updated knowledge items treated as new entries in subsequent rounds. The most recent list of knowledge items was distributed to teachers during Rounds Two and Three.

In Rounds Two and Three, teachers received the latest list of required knowledge items. They could use or disregard comments while revising decisions and were asked for qualitative feedback on labels and descriptions. Teachers described each knowledge item, identified redundancies, and suggested additions. Knowledge items were included in the final HCAP list once consensus was reached and no further changes were proposed.

Consensus criteria

There are no universally established standards for achieving consensus on every topic within each Delphi study (Keeney et al., 2006). Attaining unanimous agreement on a single survey item is generally less complex than reaching consensus on multiple items or on binary choices compared to five-point Likert scales. Consequently, the development of consensus criteria is inherently subjective and must consider the nature of the questions, the scale of the responses, and the number of participants. A systematic review by Diamond et al. (2014) found that only approximately half of Delphi studies specified and reported an explicit agreement threshold; among those that did, reported levels varied from 51 percent to 95 percent. In alignment with previous research (Ahmadi et al., 2023; Teixeira et al., 2020), this study adopted a 75 percent agreement threshold. Upon completion of three rounds, contentious activities identified as redundant by some teachers were compiled, and participants were asked to indicate which activities should be removed. Items endorsed for removal by more than half of the teachers were subsequently excluded.

Results

Round one

The Delphi Round One survey was completed by all participants. Seven of the 20 knowledge items on the initial list came to a consensus, see Table 1. They are data fundamentals (#4), assessment redesign (#11), interaction flow designer (#16), group dynamics collaboration (#19), data privacy (#21), societal impact (#23), and inclusive design (#25). We proposed two new knowledge items: tool integration (#5), and usage protocols (#22). We renamed X knowledge items: from epistemic vigilance and critical validation to critical validation (#6); from contextualization and alignment to contextualization (#8); scaffolded collaborative design to scaffolded collaboration (#12); from differentiation and personalization to personalization (#13). Then we sent the 15 activities that failed to come to an agreement to the participants for more comments in Round Two.

Round two

The Delphi Round Two survey was completed by all participants. Four of the 13 knowledge items on the Round Two list came to a consensus, see Table 1. They are tool integration (#5), contextualization (#8), personalization (#13), and usage protocols (#22). We proposed one new knowledge item: well-being advocacy (#24). We split critical dialogue and interdependence (#17) into two items: critical dialogue (#17) and critical interdependence (#18). We renamed six knowledge items: from prompt engineering and interaction to prompt engineering (#1); from bias and inaccuracy detection to bias detection (#7); understanding of AI's disciplinary approach to disciplinary methodological critique (#9); from metacognitive orchestration to reflective practice (#10) from scaffolded collaboration to scaffolded co-agency design (#12); bias mitigation and fairness protocols to equity auditing (#20). Subsequently, we forwarded the 13 knowledge items that did not reach consensus to the teachers for further feedback during Round Three.

Round three

All the teachers completed the Round Three survey. Of the remaining activities, nine reached consensus, see Table 1. They are prompt engineering (#1), disciplinary methodological critique (#9), bias detection (#7),

reflective practice (#10), scaffolded co-agency design (#12), critical dialogue (#17), critical interdependence (#18), bias auditing (#20), well-being advocacy (#24). We proposed one new item ethical integration (#14). We renamed four knowledge items: from algorithmic awareness and model limitation to limitation awareness (#2); system auditing and evaluation to system auditing (#3); from critical validation to source triangulation (#6); from role definition and task allocation to dynamic role allocation (#15). We sent the five items for final survey, results in reaching consensus. Overall, we have 25 knowledge items on the final list, see [Table 2](#).

Discussions

Mapping knowledge items onto various literacies

In this study, the proposed Human-Centric Artificial Intelligence Pedagogy (HCAP) framework, consisting of five domains, outlines the knowledge required for teachers to integrate AI effectively, ethically, and healthily. This knowledge does not exist in isolation but is fundamentally built upon a foundation of various literacies (Heung et al., 2026; Zhou, Petri, et al., 2025).

The AI-Technological Knowledge (I-TK) domain is anchored in data literacy and computational literacy, aligning with the technical skills of controlling and manipulating the AI environment (Zhou, Petri, et al., 2025). For example, data fundamentals (#4) is a direct application of data literacy, requiring an understanding of how data quality and diversity shape AI models (Heung et al., 2026). Limitation awareness (#2) and system auditing (#3) demand a form of computational and algorithmic literacy, enabling teachers to deconstruct the AI's operational logic and evaluate its reliability. Prompt engineering (#1) can be viewed as a specialized form of computational communication, a literacy for strategically interacting with AI systems to achieve desired outcomes (Celik et al., 2025).

The AI-Content Knowledge (I-CK) and AI-Pedagogical Knowledge (I-PK) domains bridge the technical and practical skills, heavily relying on critical literacies / thinking (Dilek et al., 2025). In I-CK, source triangulation (#6) and bias detection (#7) are core exercises in algorithmic literacy and media literacy, empowering teachers to function as critical validators of AI-generated content (Heung et al., 2026; Zhou, Petri, et al., 2025). Disciplinary methodological critique (#9) requires scientific literacy to compare the AI's statistical approach with human, discipline-specific modes of inquiry. Moving to I-PK, reflective practice (#10) and assessment redesign (#11) foster metacognition, a key aspect of learning literacy, while personalization (#13) leverages digital literacy to skilfully manage AI tools for tailoring education (Zhang & Zhang, 2024).

The Human-AI Collaboration Knowledge (HAIC-K) domain focuses on developing practical skills, particularly in communication and building shared understanding. Language literacy becomes paramount (Heung et al., 2026). Critical dialogue (#17) involves guiding students to engage in nuanced conversation with AI, and dynamic role allocation (#15) requires clear communication to define the AI's part in the learning process. Group dynamics collaboration (#19) draws on social and emotional learning to understand how an AI agent alters team interactions, ensuring the technology supports rather than hinders human collaboration.

Finally, the Ethical Knowledge (Ethics-K) domain focuses on critical self-awareness and societal impact. This domain is the primary vessel for civic literacy (Biagini, 2025; Heung et al., 2026). Items like societal impact (#23), well-being advocacy (#24), and inclusive design (#25) require teachers to foster an understanding of AI's role in society and a commitment to justice and equity. Equity auditing (#20) is a direct application of advanced algorithmic literacy, moving beyond detection to active mitigation of embedded biases. Usage protocols (#22) establish the normative frameworks for responsible digital citizenship in an AI-saturated world.

Overall, the HCAP framework brings the interdisciplinary AI literacy and competency (Chiu, 2025) to life in a practical way. Key literacies like data, digital, and mathematical knowledge make up the foundation of I-TK and I-PK. Meanwhile, algorithmic, media, and civic literacies give the framework its critical and emancipatory edge (Heung et al., 2026), shaping the I-CK and Ethics-K areas. This mapping shows that preparing teachers for the era of AI is not just about picking up one new skill. It is about purposefully building and connecting a diverse set of skills and literacies, both old (e.g. TPACK) and new.

Interdependence of the five domains

The five domains of the HCAP framework form a dynamic, interdependent competence rather than a set of isolated competencies (Chiu, 2025). Their synergy is what enables teachers to move from merely using AI to

Table 2. Knowledge items within HCAP after three rounds.

Knowledge items		Descriptions
Intelligent Technological Knowledge (I-TK): Mastery of AI systems' operational logic, capabilities, and limitations		
#1	Prompt engineering	The ability to craft, iterate, and refine strategic prompts to reliably generate high-quality, relevant, and pedagogically useful outputs from generative AI models.
#2	Limitation awareness	The knowledge of fundamental principles such as training data and machine learning. It is important to understand the reasons why AI models may exhibit biases, generate hallucinations, or make errors, as well as to recognize the specific functions and limitations of widely used educational AI platforms. This knowledge forms the basis of algorithmic literacy.
#3	System auditing	The skill to critically evaluate an AI tool's reliability, data security, privacy policies, and embedded ethical considerations before deployment in the classroom.
#4	Data fundamentals	The knowledge of how data quality, quantity, and diversity shape AI model behavior, and of basic data processing and interpretation in the context of AI outputs.
#5	Tool integration	The knowledge of how to effectively integrate multiple AI tools, such as a text generator, an image creator, and a data analyzer, into a seamless workflow to accomplish complex teaching and learning tasks.
Intelligent Content Knowledge (I-CK): The ability to critically curate, validate, and contextualize AI-generated disciplinary content.		
#6	Source triangulation	The ability to cross-reference AI-generated information against authoritative, human-curated sources to verify factual accuracy, conceptual soundness, and argumentative validity. This is associated with epistemic vigilance.
#7	Bias detection	The skill to identify and analyze potential algorithmic biases, cultural stereotypes, and representational gaps in AI-generated content that may arise from its training data.
#8	Contextualization	The ability to adapt, frame, and align AI-generated content with specific curriculum standards, learning objectives, and the prior knowledge of a diverse student body.
#9	Disciplinary methodological critique	The knowledge of how an AI's approach to solving problems or generating content (e.g. statistical pattern-matching) may differ from human, disciplinary-specific methods of inquiry and reasoning (e.g. history and experience).
Intelligent Pedagogical Knowledge (I-PK): The strategic design of AI-augmented learning experiences.		
#10	Reflective practice	The ability to design learning activities that explicitly prompt students to reflect on how they are using AI, how it influences their thinking process, and what they are learning from the human-AI interaction.
#11	Assessment redesign	The ability to create novel assessment methods that evaluate the process of collaborating with AI, such as assessing a student's ability to refine prompts, critique AI outputs, and integrate them into original work, rather than just the final product.
#12	Scaffolded co-agency design	The ability to structure lessons that define clear, dynamic, and complementary roles for both the student and the AI agent to foster productive collaboration.
#13	Personalization	The skill to leverage AI tools to efficiently create a range of learning materials, provide targeted feedback, and offer adaptive learning paths that cater to varied student readiness levels, interests, and learning profiles.
#14	Ethical integration	The ability to proactively incorporating ethical considerations, such as disclosure requirements, data privacy discussions, and bias assessments, into lesson plans. The plans demonstrate a commitment to responsible education.
Human-AI Collaboration Knowledge (HAIC-K): Structuring effective interactions and defining roles for AI within learning.		
#15	Dynamic role allocation	The knowledge to define and assign context-specific, assign roles to AI in a learning activity, such as a questioner, a simulation partner, a data visualizer, or a silent procedural assistant.
#16	Interaction Flow designer	The skill to design the temporal sequence and rules of engagement for human-AI interaction, including when to introduce the AI, when to pause its use for individual or group reflection, and how to synthesize its contributions.
#17	Critical dialogue	The knowledge to guide and facilitate students to engage in a critical, iterative dialogue with AI systems, teaching them to probe assumptions, challenge conclusions, and use the AI as a tool for deepening inquiry rather than for finding quick answers.
#18	Critical interdependence	The ability to cultivate and foster a classroom culture where the value of human judgment, creativity, and ethics is elevated, and AI is positioned as a powerful but subordinate tool that augments, rather than replaces, human intellect.
#19	Group dynamics collaboration	The knowledge of how the introduction of an AI agent affects peer-to-peer collaboration, communication, and the distribution of cognitive labor within a student team.
Ethical Knowledge (Ethics-K): The normative frameworks for responsible and equitable AI use in education.		
#20	Equity auditing	The commitment and skill to proactively audit AI tools and generated content for biases related to race, gender, socioeconomic status, and culture, and to implement strategies to mitigate their impact on students.
#21	Data privacy	The knowledge of data protection regulations, ensuring student data is not exploited, and empowering students with knowledge about their digital footprint and rights when interacting with AI.
#22	Usage protocols	The ability to establish clear classroom policies for disclosing AI use, maintaining academic integrity, and defining ultimate human accountability for decisions informed by AI.
#23	Societal impact	The ability to guide students to critically examine the broader societal, economic, and environmental implications of AI systems, including issues of job displacement, misinformation, and energy consumption.
#24	Well-being advocacy	The commitment and skill to promote the healthy, balanced use of AI to prevent over-reliance, protect student agency and self-efficacy, and mitigate potential negative impacts on mental health and creative confidence.
#25	Inclusive design	The ability to ensure that the use of AI tools in the classroom adheres to principles of universal design for learning and does not create new barriers or exacerbate existing inequities for students with disabilities.

orchestrating human-AI learning partnerships. This interaction is fundamentally powered by a teacher's integrated command of various literacies.

The process begins with I-TK, the foundational engine of the framework. This domain is rooted in data and computational literacy (Heung et al., 2026; Zhou, Petri, et al., 2025). Before designing the lesson, the teacher must audit potential AI tools (system auditing, #3), evaluating their data privacy policies and reliability. They use their understanding of data fundamentals (#4) to know that AI's knowledge is limited to its training data, leading directly to limitation awareness (#2). This technical foundation serves as a critical prerequisite for all subsequent activities. For example, teachers' skill in prompt engineering (#1) directly determines the quality of AI output (raw materials) that students and teachers will later critique (Celik et al., 2025).

When this AI output is produced, it instantly triggers the I-CK domain, making teachers' algorithmic literacy and scientific literacy essential (Dilek et al., 2025; Heung et al., 2026; Moylan & Code, 2024). The AI-generated outputs are accepted uncritically. Then teachers guide students to employ source triangulation (#6), cross-referencing the outputs against reputable sources. Teachers and students engage in bias detection (#7). Crucially, they apply disciplinary methodological critique (#9), questioning whether the outputs fit the disciplinary content. The effectiveness of this critical validation in I-CK is dependent on the quality of the initial prompt from I-TK; an inadequate prompt results in outputs that are challenging and time-consuming to validate.

With a critically vetted AI-generated resource in hand, the teacher then applies I-PK to design the learning experience. This is where digital literacy and pedagogical knowledge merge (Heung et al., 2026; Zhang & Zhang, 2024). The teacher uses personalization (#13) to leverage AI as a tool for differentiation, perhaps generating simplified explanations for some students and more complex ethical dilemmas for others (Mulaudzi & Hamilton, 2025). They redesign assessment (#11), for example, shifting the focus from a summative report (learning as product) to a process portfolio (learning as process) that includes the students' prompts, their critique of the AI's outputs, and a reflection. This portfolio is a product of scaffolded co-agency design (#12). Teachers structure a clear workflow; the AI acts as an initial advisory technology; students as critical analysts and synthesizers. Furthermore, the teacher proactively plans for ethical integration (#14), deciding to explicitly discuss data privacy with students before they even interact with the AI tool (Aldemir et al., 2025).

This pedagogical design creates the blueprint for interaction, which is realized through HAIC-K. Teachers implement the dynamic role allocation (#15) designed in I-PK by suggesting the roles of students and AI in the learning experiences. Teachers then facilitate critical dialogue (#17), coaching students to critically evaluate the outputs. This structured interaction fosters critical interdependence (#18). In this case, AI is seen as advisory intelligence, rather than replaces human judgment and empathy. Teachers also monitor group dynamics collaboration (#19), observing how the introduction of AI affects student discussion and ensuring it promotes rather than hinders collaboration.

Ethics-K serves as the framework's moral compass and critical conscience, providing ethical guidance throughout the process. This domain is the primary manifestation of civic literacy (Heung et al., 2026). Teachers' actions in all other domains are guided by Ethics-K. System auditing (I-TK, #3) was driven by data privacy (Ethics-K, #21) concerns. Bias detection (I-CK, #7) is part of a broader commitment to bias auditing (Ethics-K, #20). The pedagogical design includes usage protocols (Ethics-K, #22), such as requiring students to disclose their AI use. Teachers should embrace societal impact (Ethics-K, #23) and inclusive design (Ethics-K, #25) to promote educational equity. Finally, the teacher acts as a well-being advocate (Ethics-K, #24), ensuring students and teachers have happy and healthy learning experiences with AI.

In summary, the five domains of HCAP are fundamentally interconnected, forming a dynamic and constructive cycle (Chiu, 2025). I-TK supplies foundational knowledge and capability; I-CK offers an essential evaluative framework; I-PK shapes effective instructional strategies; HAIC-K facilitates meaningful real-time engagement; and Ethics-K upholds ethical standards to foster responsible, equitable, and empowering educational experiences. Deficiency in any domain can undermine the integrity of the entire system. For example, educators with strong proficiency in I-TK but limited competence in Ethics-K may produce technically adept yet ethically uninformed students, while those proficient in Ethics-K but lacking strength in I-CK might accept erroneous AI outputs uncritically.

Practical use of the HCAP

We suggest two ways of using HCAP to improve teacher education. The first suggestion is to develop a competency-based assessment portfolio (Chang & Hwang, 2023). Assessment of HCAP proficiency should mirror its practical, integrated nature. Instead of using written assignments, teachers would build a human-AI teaching portfolio. This portfolio would contain concrete artifacts demonstrating mastery of the 25 knowledge items across the five domains. Evidence could include: video recordings of themselves facilitating a critical dialogue (#17) between students and an AI; samples of student work with annotated feedback showing how they used assessment redesign (#11) to evaluate the collaboration process; a reflective journal detailing their system auditing (#3) of a new AI tool and the resulting usage protocols (#22) they established for their classroom; and a curated collection of prompts they engineered (prompt engineering, #1) and the subsequent, critically validated outputs (source triangulation, #6). This portfolio would provide a holistic and evidence-based measure of a teacher's ability to apply HCAP in practice, highlighting their growth and areas for further development.

The second one is to foster collaborative, cross-disciplinary professional learning communities. The complexity of HCAP necessitates breaking down the traditional isolation of teacher education. Teacher education programs should establish professional learning communities (Cai et al., 2025; Chiu, 2022) where pre-service and in-service teachers, technology specialists, and subject-matter experts collaborate. In these communities, a math teacher struggling with how to implement disciplinary methodological critique (#9) for AI-generated outputs could partner with a computer science teacher who has strong algorithmic literacy. A group could collaboratively develop a cross-curricular project, requiring them to negotiate and align their ethical integration (#14) strategies and scaffolded co-agency designs (#12). These communities become environments for shared best practices, where teachers can safely experiment, discuss challenges like equity auditing (#20). These collectively develop a robust understanding of how HCAP manifests differently across various subjects and grade levels. This collaborative model ensures that the development of HCAP is a social, ongoing process, rather than a static set of individual competencies.

More discussions on HAIC-K and ethics-K in the HCAP framework

The central idea of HCAP framework emerges from the interaction of its domains, with HAIC-K and Ethics-K ensuring these interactions remain human-centric. The two newly added knowledge domains, i.e. HAIC-K and Ethics-K, have dynamic relationships with I-TK, I-CK, and I-PK to form a robust framework for responsible and effective teaching practice using AI. Table 3 presents a matrix along with descriptions that explain how the five domains work together to form a comprehensive model for integrating AI into teaching.

The relationship between Ethics-K and I-TK involves an important exchange of challenges and insights. While technical skills enable people to use AI tools, using these tools reveals their biases and limitations. This, in turn, helps deepen and reinforce ethical understanding. For example, a teacher using an AI image generator (I-TK) might notice it only produces stereotypes, an observation made through an ethical lens (Ethics-K). This newfound ethical insight then directly constrains and refines future technical applications, pushing teachers to use more inclusive prompts. Thus, Ethics-K ensures that I-TK is not applied indiscriminately but is guided by a commitment to responsibility and fairness. This creates a feedback loop where technical exploration heightens ethical vigilance, and ethical principles shape technical proficiency.

Similarly, HAIC-K and I-TK share a symbiotic relationship of facilitation and demand. HAIC-K, which involves designing collaborative roles for AI, relies entirely on TK for its execution. For example, teachers use AI as an opponent in a classroom debate (HAIC-K); students need to have sophisticated prompt engineering skills (I-TK) for meaningful debate. In this sense, I-TK facilitates HAIC-K. Conversely, the collaborative goals defined by HK create a direct demand for specific, advanced technical skills. The vision for human-AI collaborations drives the need for technical mastery, ensuring TK development is purposeful and directly serving human-centered learning.

When Ethics-K interacts with I-CK, it establishes an essential critical framework. Ethical principles provide the lens for validating AI-generated content, ensuring it is not just accurate but also equitable and representative. The act of applying an ethical protocol (Ethics-K) to an AI-generated historical narrative (I-CK) can reveal Eurocentric biases, turning content validation into a profound teachable moment about algorithmic

Table 3. A matrix along with descriptions for how HK and EK interact with I-TK, I-CK, and I-PK.

Initiating Domain	Interacting Domain	Nature of the Interaction	Practical Example
I-TK enables I-CK	I-CK refines I-TK	Technical skills provide the raw material for critical content validation, which in turn improves technical use.	A teacher uses prompt engineering (I-TK) to generate multiple explanations of photosynthesis. They then critically evaluate (I-CK) the outputs for accuracy and identify where the AI oversimplifies the Calvin cycle, leading them to refine their prompts for greater precision.
I-TK Informs Ethics-K	Ethics-K Constrains I-TK	Technical expertise strengthens ethical awareness, while ethical understanding helps ensure technical skills are applied safely and responsibly.	A teacher uses an AI image generator (I-TK) to create visuals for a lesson on community helpers. They notice the AI only generates images of male construction workers and female teachers. Their I-TK allows them to use more specific, inclusive prompts (e.g. a diverse team of construction workers including women and people of color) to actively counteract the bias they identified through their Ethics-K lens.
I-PK structures HAIC-K	HAIC-K operationalizes I-PK	Pedagogical design creates an environment for collaboration, and the collaborative model brings pedagogy to life.	A teacher designs a project-based learning unit (I-PK) where students must solve a local environmental issue. They structure the AI's role (HAIC-K) as a data analyst and scenario simulator, while students act as decision-makers, ensuring the pedagogy is executed through structured collaboration.
I-TK facilitates HAIC-K	HAIC-K demands I-TK	Technical knowledge allows for the dynamic configuration of AI, and collaborative goals create the need for specific technical skills.	To make an AI act as a devil's advocate in a classroom debate (HAIC-K), a teacher must use sophisticated prompt engineering (I-TK). The collaborative goal drives the need for advanced technical skill.
HK enhances I-CK	I-CK guides HAIC-K	Collaboration with AI can deepen content understanding, and a firm grasp of content is needed to guide the collaboration productively.	Students collaborate with an AI (HAIC-K) to write a poem in a specific style. To evaluate and refine the AI's output, they must deeply understand the poetic form and devices (I-CK), turning the collaboration into a content mastery exercise.
Ethics-K & HAIC-K are the Core Pillars	All Domains	These two domains are interdependent and provide the essential foundation for all other interactions, ensuring they are human-centric and responsible.	In any activity involving I-TK, I-CK, and I-PK, the teacher must constantly ask: "Is this collaboration fair and effective (HAIC-K)?" and "Are we upholding our ethical commitments (Ethics-K)?" For example, using AI to analyze student data (I-TK) must be governed by privacy rules (Ethics-K) and used to empower, not replace, teacher-student interactions (HAIC-K).

justice. This process is bidirectional; the very act of scrutinizing content (I-CK) through an ethical lens can uncover new ethical dilemmas (Ethics-K), fostering a continuous cycle of critical engagement.

The interaction between HAIC-K and I-CK demonstrates how collaboration can be a powerful vehicle for deep learning. Engaging with an AI to co-create a product, such as a poem (HAIC-K), is not a passive activity. It requires a firm command of the content (I-CK) to guide the AI, evaluate its contributions, and refine the final output. The collaborative process, therefore, forces a deep engagement with the subject matter, transforming the activity from simple content consumption into an active exercise in content application and mastery.

Finally, HAIC-K and Ethics-K jointly shape and expressed through I-PK. I-PK provides the structural design for learning, such as a project-based unit, while HAIC-K operationalizes this design by defining the AI's role as a collaborator, bringing the pedagogy to life. Simultaneously, Ethics-K monitors this pedagogical implementation, ensuring it is equitable, for example, by guaranteeing all students have access to the AI tool. PK creates a stage, HAIC-K directs the interaction between actors (students and AI), and Ethics-K serves as the script supervisor, ensuring the entire production aligns with core values.

Overall, HAIC-K and Ethics-K are the indispensable core pillars of I-PTACK. HAIC-K is the active, operationalizing force that translates pedagogical plans into dynamic, collaborative experiences and drives the purposeful application of AI technical skills. Ethics-K is the pervasive moral compass that scrutinizes content, governs technical use, and ensures equity in pedagogical practice. Together, they ensure that the integration of I-TK, I-CK, and I-PK remains fundamentally human-centric, critically engaged, and ethically responsible, transforming AI from a mere tool into a accountable partner in the educational process.

Ethics-K governs I-CK I-CK reveals Ethics-K dilemmas Ethical principles provide the lens for validating content, and the act of validation uncovers new ethical issues. Before using an AI-generated historical narrative about colonial expansion (I-CK), the teacher applies an ethical protocol (Ethics-K) to check for marginalized voices. Finding a Eurocentric bias, they use this as a teachable moment about algorithmic bias.

Ethics-K monitors I-PK I-PK implements Ethics-K Ethical vigilance ensures pedagogical designs are equitable, and pedagogical strategies are the vehicle for teaching ethics. A teacher redesigns an assessment (I-PK) to include AI-generated feedback. Ethics-K ensures this is done equitably (e.g. all students have access) and that the assessment includes a reflective component on the AI's feedback, teaching students critical evaluation.

Ethics-K & HAIC-K are the Core Pillars All Domains These two domains are interdependent and provide the essential foundation for all other interactions, ensuring they are human-centric and responsible. In any activity involving I-TK, I-CK, and I-PK, the teacher must constantly ask: "Is this collaboration fair and effective (HAIC-K)?" and "Are we upholding our ethical commitments (Ethics-K)?" For example, using AI to analyze student data (I-TK) must be governed by privacy rules (Ethics-K) and used to empower, not replace, teacher-student interactions (HAIC-K).

Conclusions, limitations and future research directions

This three-round Delphi study identified and refined a consensus list of 25 essential knowledge items required for the five domains of the HCAP framework. This outcome provides a foundational and empirically grounded model for the knowledge and skills teachers need to healthily, productively and ethically integrate AI into their educational practice.

This study has four limitations that warrant attention in future research. First, the participant panel consisted solely of teachers from Hong Kong, which may limit the generalizability of the HCAP knowledge items across different cultural and educational systems. Future studies should recruit international and more diverse panels to validate and potentially expand this framework. Second, the consensus was built on expert opinion and self-reported data through the Delphi method. While robust for gathering agreement, this approach does not verify whether the identified knowledge items translate into effective teaching practices or improved student outcomes. Future research should employ observational and intervention designs to test the application of HCAP in authentic classrooms and measure its impact on teaching efficacy and student learning. Third, the initial list was drafted by a few researchers, which may influence the participants' ideas. Future studies should further verify the findings. Finally, the study does not address the potential variance in HCAP requirements across different subjects, student age groups, or specific AI tools. The knowledge needed by a mathematics and science teacher using AI for lab simulations may differ from that of a language arts teacher using it for essay coaching. Future research should investigate these nuances to develop more tailored, subject – and level-specific HCAP guidelines.

Author contributions

Thomas K.F. CHIU: Conceptualization, Methodology, Data Collection, Formal analysis, Funding acquisition, Validation, Writing – Original Draft, Writing – Review & Editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work is supported by Hong Kong Research Council – General Research Fund [grant number 14613024] and The Chinese University of Hong Kong – Teaching Development and Language Enhancement [Grant number 4171207]. University Grants Committee.

Consent for publication

We have consented to the submission of the manuscript to journals.

Ethics approval and consent to participate

This study got ethical clearance from the corresponding author's university and got the consents from the participants to collect their data

Data availability statement

The datasets used for the current study are available from the corresponding author on reasonable request.

ORCID

Thomas K.F. Chiu  <http://orcid.org/0000-0003-2887-5477>

References

- Ahmadi, A., Noetel, M., Parker, P. D., Ryan, R., Ntoumanis, N., Reeve, J., Beauchamp, M., Dicke, T., Yeung, A., Ahmadi, M., Bartholomew, K., Chiu, T. K. F., Curran, T., Erturan, G., Flunger, B., Frederick, C., Froiland, J. M., González-Cutre, D., Haerens, L., & Lonsdale, C. (2023). A classification system for teachers' motivational behaviours recommended in self-determination theory interventions. *Journal of Education Psychology*, 115(8), 1158–1176. <https://doi.org/10.1037/edu0000783>
- Aldemir, T., Bicer, A., Kilinc, S., Moon, J., & Kwok, M. (2025). Exploring emergent AHCAP competencies in a two-week AI literacy module for preservice teachers. *Teaching and Teacher Education*, 168, 105231. <https://doi.org/10.1016/j.tate.2025.105231>
- Biagini, G. (2025). Towards an AI-literate future: A systematic literature review exploring education, ethics, and applications. *International Journal of Artificial Intelligence in Education*, 35(4), 2616–2666. <https://doi.org/10.1007/s40593-025-00466-w>
- Cai, H., Han, B., Lu, L., Wong, L. H., & Gu, X. (2025). When AI joins the conversation: The impact of AI-powered teacher dashboard on pre-service teachers' collaborative interaction in the video-based professional learning communities. *Interactive Learning Environments*. <https://doi.org/10.1080/10494820.2025.2546642>
- Celik, I. (2023). Towards Intelligent-TPACK: An empirical study on teachers' professional knowledge to ethically integrate artificial intelligence (AI)-based tools into education. *Computers in Human Behavior*, 138, 107468. <https://doi.org/10.1016/j.chb.2022.107468>
- Celik, I., Kontkanen, S., Laru, J., & Dalyanci, A. A. (2025). Co-constructing adaptive lesson plans with GenAI: Pre-service teachers' intelligent-TPACK and prompt engineering strategies. *Computers & Education*, 241, 105485. <https://doi.org/10.1016/j.compedu.2025.105485>
- Chang, C. C., & Hwang, G. J. (2023). An experiential learning-based virtual reality approach to fostering problem-solving competence in professional training. *Interactive Learning Environments*, 31(8), 4713–4728. <https://doi.org/10.1080/10494820.2021.1979049>
- Chiu, T. K. F. (2022). School learning support for teacher technology integration from a self-determination theory perspective. *Educational Technology Research and Development*, 70(3), 931–949. <https://doi.org/10.1007/s11423-022-10096-x>
- Chiu, T. K. F. (2024). The impact of generative AI (GenAI) on practices, policies and research direction in education: A case of ChatGPT and Midjourney. *Interactive Learning Environments*, 32(10), 6187–6203. <https://doi.org/10.1080/10494820.2023.2253861>
- Chiu, T. K. F. (2025). Developing intelligent-TPACK (HCAP) framework from unpacking AI literacy and competency: Implementation strategies and future research direction. *Interactive Learning Environments*, 33(7), 4189–4192. <https://doi.org/10.1080/10494820.2025.2545053>
- Chiu, T. K. F., Ahmand, Z., & Çoban, M. (2024). Development and validation of teacher artificial intelligence (AI) competence self-efficacy (TAICS) scale. *Education and Information Technologies*, 30(5), 6667–6685. <https://doi.org/10.1007/s10639-024-13094-z>
- Chiu, T. K. F., & Rospigliosi, P. A. (2025). Encouraging human-AI collaboration in interactive learning environments. *Interactive Learning Environments*, 33(2), 921–924. <https://doi.org/10.1080/10494820.2025.2471199>
- Diamond, I. R., Grant, R. C., Feldman, B. M., Pencharz, P. B., Ling, S. C., Moore, A. M., & Wales, P. W. (2014). Defining consensus: A systematic review recommends methodologic criteria for reporting of Delphi studies. *Journal of Clinical Epidemiology*, 67(4), 401–409. <https://doi.org/10.1016/j.jclinepi.2013.12.002>

- Dilek, M., Baran, E., & Aleman, E. (2025). AI literacy in teacher education: Empowering educators through critical co-discovery. *Journal of Teacher Education*, 76(3), 294–311. <https://doi.org/10.1177/00224871251325083>
- Heung, Y. M. E., Yin, H. B., English, L., & Chiu, T. K. F. (2026). Theorizing AI literacy development using habermas' three cognitive knowledge interests: A STEM interdisciplinary perspective. *Computers & Education*, 242, 105492. <https://doi.org/10.1016/j.compedu.2025.105492>
- Hughes, L., Dwivedi, Y. K., Malik, T., Shawosh, M., Albashrawi, M. A., Jeon, I., Dutot, V., Appanderanda, M., Crick, T., De', R., Fenwick, M., Gunaratnege, S. M., Jurcys, P., Kar, A. K., Kshetri, N., Li, K., Mutasa, S., Samothrakis, S., Wade, M., & Walton, P. (2025). AI agents and agentic systems: A multi-expert analysis. *Journal of Computer Information Systems*, 65(4), 489–517. <https://doi.org/10.1080/08874417.2025.2483832>
- Jorm, A. F. (2015). Using the Delphi expert consensus method in mental health research. *The Australian and New Zealand Journal of Psychiatry*, 49(10), 887–897. <https://doi.org/10.1177/0004867415600891>
- Karataş, F., & Ataç, B. A. (2025). When TPACK meets artificial intelligence: Analyzing TPACK and AI-TPACK components through structural equation modelling. *Education and Information Technologies*, 30(7), 8979–9004. <https://doi.org/10.1007/s10639-024-13164-2>
- Keeney, S., Hasson, F., & McKenna, H. (2006). Consulting the oracle: Ten lessons from using the delphi technique in nursing research. *Journal of Advanced Nursing*, 53(2), 205–212. <https://doi.org/10.1111/j.1365-2648.2006.03716.x>
- Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–16. <https://doi.org/10.1145/3313831.3376727>
- Mishra, P., & Koehler, M. J. (2006). Technological pedagogical content knowledge: A framework for integrating technology in teachers' knowledge. *Teachers College Record: The Voice of Scholarship in Education*, 108(6), 1017–1054. <https://doi.org/10.1111/j.1467-9620.2006.00684.x>
- Mishra, P., Warr, M., & Islam, R. (2023). TPACK in the age of ChatGPT and generative AI. *Journal of Digital Learning in Teacher Education*, 39(4), 235–251. <https://doi.org/10.1080/21532974.2023.2247480>
- Moylan, R., & Code, J. (2024). Algorithmic futures: An analysis of teacher professional digital competence frameworks through an algorithm literacy lens. *Teachers and Teaching*, 30(4), 452–470. <https://doi.org/10.1080/13540602.2023.2263732>
- Mulaudzi, L. V., & Hamilton, J. (2025). Lecturer's perspective on the role of AI in personalized learning: Benefits, challenges, and ethical considerations in higher education. *Journal of Academic Ethics*, 23(4), 1571–1591. <https://doi.org/10.1007/s10805-025-09615-1>
- Pamuk, S. (2012). Understanding preservice teachers' technology use through TPACK framework. *Journal of Computer Assisted Learning*, 28(5), 425–439. <https://doi.org/10.1111/j.1365-2729.2011.00447.x>
- Teixeira, P. J., Marques, M. M., Silva, M. N., Brunet, J., Duda, J., Haerens, L., La Guardia, J., Lindwall, M., Lonsdale, C., Markland, D., & Others (2020). A classification of motivation and behavior change techniques used in self-determination theory-based interventions in health contexts. *Motivation Science*, 6(4), 438–455. <https://doi.org/10.1037/mot0000172>
- UNESCO. (2024). *AI competency framework for teachers*. United Nations Educational, Scientific and Cultural Organization. <https://www.unesco.org/en/articles/ai-competency-framework-teachers>
- Vučković, A., & Sikimić, V. (2025). Global justice and the use of AI in education: Ethical and epistemic aspects. *AI & Society*, 40(4), 3087–3104. <https://doi.org/10.1007/s00146-024-02076-x>
- Wen, Y., Wang, Z., & Guo, X. (2025). Trends and applications of AI in immersive learning environments: A systematic review of empirical research. *Interactive Learning Environments*. <https://doi.org/10.1080/10494820.2025.2524029>
- Wong, J. M. S. (2025). Speculative futures of education: Utopian and dystopian scenarios envisioned by ChatGPT, gemini, and deepseek. *Discover Education*, 4(1), 261. <https://doi.org/10.1007/s44217-025-00692-3>
- Zhang, J., & Zhang, Z. (2024). AI in teacher education: Unlocking new dimensions in teaching support, inclusive learning, and digital literacy. *Journal of Computer Assisted Learning*, 40(4), 1871–1885. <https://doi.org/10.1111/jcal.12988>
- Zhou, C., Petri, S. M. R., & Bhattacharyya, S. (2025). AI literacy and professional development in STEM education. In S. Papadakis (Ed.), *AI roles and responsibilities in education. Signals and communication technology*. Springer. https://doi.org/10.1007/978-3-031-96855-6_3
- Zhou, X., Li, Y., Chai, C. S., & Chiu, T. K. F. (2025). Defining, enhancing, and assessing artificial intelligence literacy and competency in K-12 education from a systematic review. *Interactive Learning Environments*, 33(10), 5766–5788. <https://doi.org/10.1080/10494820.2025.2487538>